

Homework 2 cahpter3

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3.7 Exercises

Do exercises 3.1, 3.2, 3.3, 3.4, 3.5, 3.7, 3.8 and 3.9 from the online Hyndman book. Please include your Rpubs link along with.pdf file of your run code

3.1

Consider the GDP information in `global_economy`. Plot the GDP per capita for each country over time. Which country has the highest GDP per capita? How has this changed over time?

```
library(tidyverse)
```

```
Warning: package 'ggplot2' was built under R version 4.5.2
```

```
Warning: package 'tibble' was built under R version 4.5.2
```

```
Warning: package 'tidyr' was built under R version 4.5.2
```

```
Warning: package 'readr' was built under R version 4.5.2
```

```
Warning: package 'purrr' was built under R version 4.5.2
```

```
Warning: package 'dplyr' was built under R version 4.5.2
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.0      v readr      2.1.6
v forcats    1.0.1      v stringr    1.6.0
v ggplot2     4.0.2      v tibble     3.3.1
v lubridate  1.9.4      v tidyr      1.3.2
v purrr       1.2.1
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(dplyr)
library(ggplot2)
#install.packages("fpp3")
library(ggtime)
```

Warning: package 'ggtime' was built under R version 4.5.2

```
library(fpp3)
```

Warning: package 'fpp3' was built under R version 4.5.2

```
-- Attaching packages ----- fpp3 1.0.3 --
v tsibble      1.2.0      v feasts      0.5.0
v tsibbledata  0.4.1      v fable       0.5.0
```

Warning: package 'tsibble' was built under R version 4.5.2

Warning: package 'feasts' was built under R version 4.5.2

Warning: package 'fabletools' was built under R version 4.5.2

Warning: package 'fable' was built under R version 4.5.2

```
-- Conflicts ----- fpp3_conflicts --
x lubridate::date() masks base::date()
x dplyr::filter()   masks stats::filter()
x tsibble::intersect() masks base::intersect()
x tsibble::interval() masks lubridate::interval()
x dplyr::lag()       masks stats::lag()
x tsibble::setdiff() masks base::setdiff()
x tsibble::union()   masks base::union()
```

since we have a large datasets it is safe to find the highest GDP per capita for every year, then observe their changes over time by using plot.

```
global_economy |>
  mutate(gdp_per_capita = GDP/Population ) |>
  filter(!is.na(gdp_per_capita)) |>
  as_tibble() |>
  group_by(Year) |>
  slice_max(
    order_by = gdp_per_capita,
    n = 1,
    with_ties = FALSE
  ) |>
  select(Year, Country, gdp_per_capita)
```

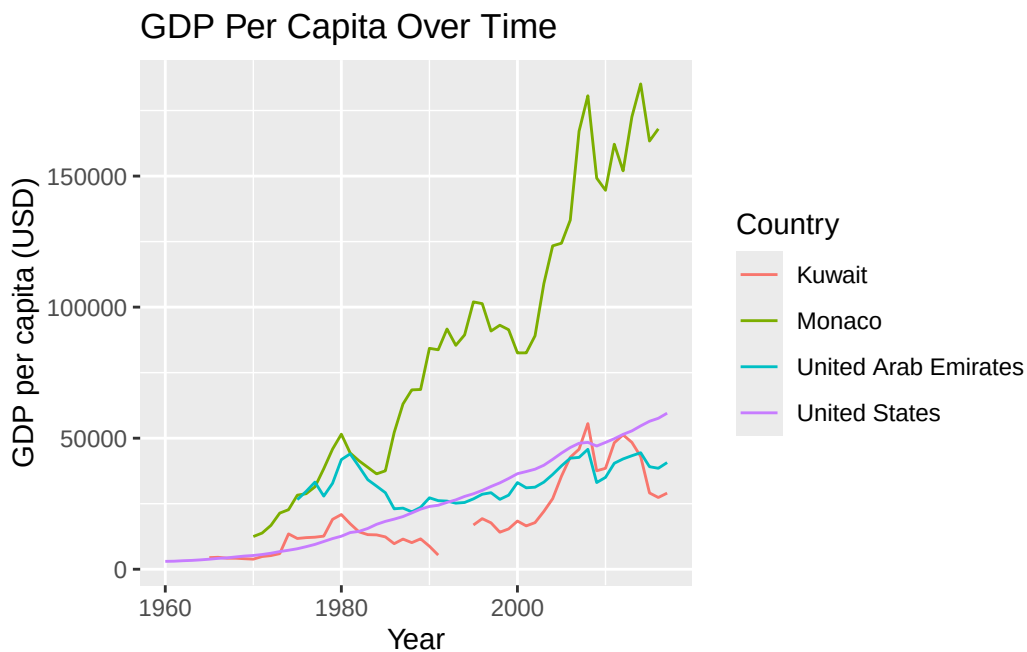
```
# A tibble: 58 x 3
# Groups:   Year [58]
   Year Country      gdp_per_capita
  <dbl> <fct>         <dbl>
1  1960 United States    3007.
2  1961 United States    3067.
3  1962 United States    3244.
4  1963 United States    3375.
5  1964 United States    3574.
6  1965 Kuwait          4429.
7  1966 Kuwait          4556.
8  1967 United States    4336.
9  1968 United States    4696.
10 1969 United States    5032.
# i 48 more rows
```

```
global_economy |>
  mutate(gdp_per_capita = GDP / Population) |>
  filter(
    Country %in% c(
      "United States",
      "Kuwait",
      "Monaco",
      "United Arab Emirates"
    )
  ) |>
  ggplot(aes(
```

```

    x = Year,
    y = gdp_per_capita,
    color = Country
  )) +
  geom_line(na.rm = TRUE) +
  labs(
    title = "GDP Per Capita Over Time",
    x = "Year",
    y = "GDP per capita (USD)",
    color = "Country"
  )

```



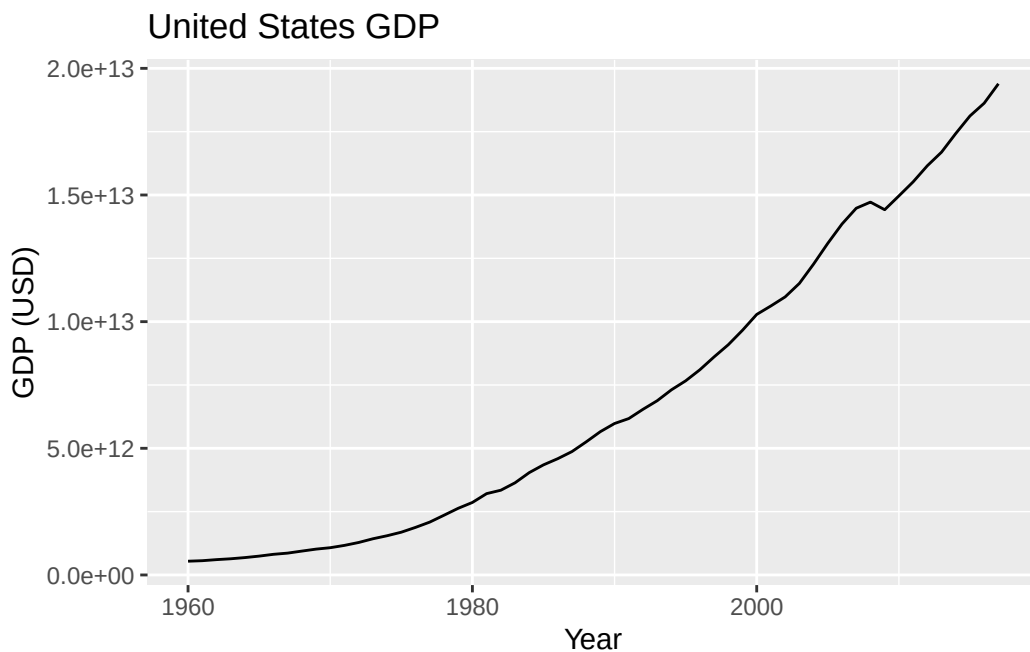
##Interpretation

The country with the highest GDP per capita for most of the period is Monaco. Monaco shows a strong upward trend over time, although there are some periods of decline, especially around 1980's, 2000 and 2010. Overall, Monaco's GDP per capita increased significantly over time. The United Arab Emirates, the United States, and Kuwait also show an overall upward trend. The United States has a more stable trend, while Kuwait, Monaco, and the United Arab Emirates show more fluctuations and several periods of decline.

3.2 .For each of the following series, make a graph of the data. If transforming seems appropriate, do so and describe the effect.

- United States GDP from `global_economy`.

```
global_economy |>
  filter(Country == "United States") |>
  autoplot(GDP) +
  labs(title = "United States GDP",
       x = "Year",
       y = "GDP (USD)")
```

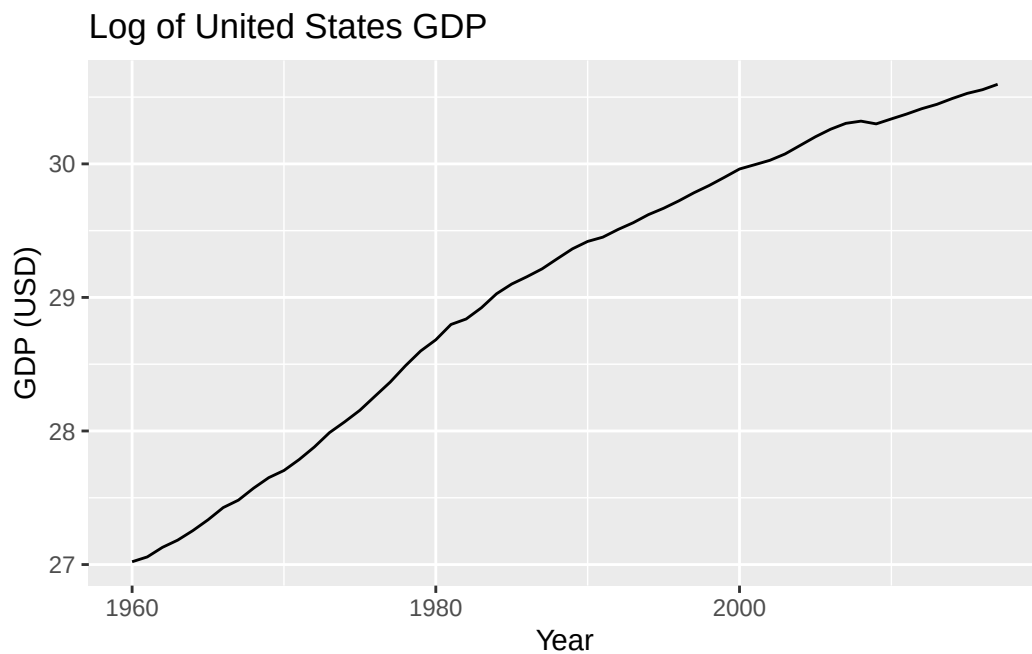


Interpretation

The original U.S. GDP series shows a strong upward trend, with the magnitude of the changes increasing as GDP increases. The log transformation compresses the larger values and makes the growth pattern more nearly linear, helping to stabilize the variation over time.

```
global_economy |>
  filter(Country == "United States") |>
  autoplot(log(GDP)) +
  labs(title = "Log of United States GDP",
```

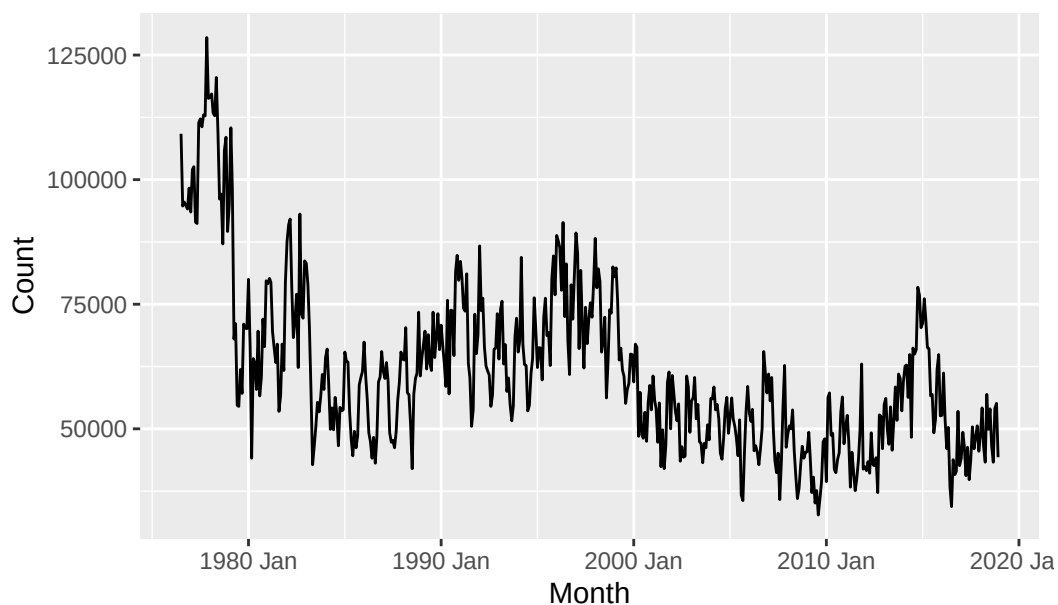
```
x = "Year",  
y = "GDP (USD)"
```



- Slaughter of Victorian “Bulls, bullocks and steers” in `aus_livestock`.

```
aus_livestock |>  
  filter(State == "Victoria",  
         Animal == "Bulls, bullocks and steers") |>  
  ggplot(aes(x = Month, y = Count)) +  
  geom_line() +  
  labs(title = "Slaughter of Victorian Bulls, bullocks and steers", x = "Month", y = "Count")
```

Slaughter of Victorian Bulls, bullocks and steers



```
victoria_bulls <- aus_livestock |>
  filter(
    State == "Victoria",
    Animal == "Bulls, bullocks and steers"
  )

dcmp <- victoria_bulls |>
  model(stl = STL(Count))

components(dcmp)
```

```
# A dable: 510 x 9 [1M]
# Key:   Animal, State, .model [1]
# :      Count = trend + season_year + remainder
```

Animal	State	.model	Month	Count	trend	season_year	remainder
<fct>	<fct>	<chr>	<mth>	<dbl>	<dbl>	<dbl>	<dbl>
1 Bulls, bullocks an~	Vict~	stl	1976 Jul	109200	9.51e4	-1381.	15518.
2 Bulls, bullocks an~	Vict~	stl	1976 Aug	94700	9.58e4	-6576.	5518.
3 Bulls, bullocks an~	Vict~	stl	1976 Sep	95500	9.65e4	-5098.	4144.
4 Bulls, bullocks an~	Vict~	stl	1976 Oct	94800	9.71e4	1347.	-
3695.							
5 Bulls, bullocks an~	Vict~	stl	1976 Nov	94100	9.81e4	3241.	-
7203.							

```

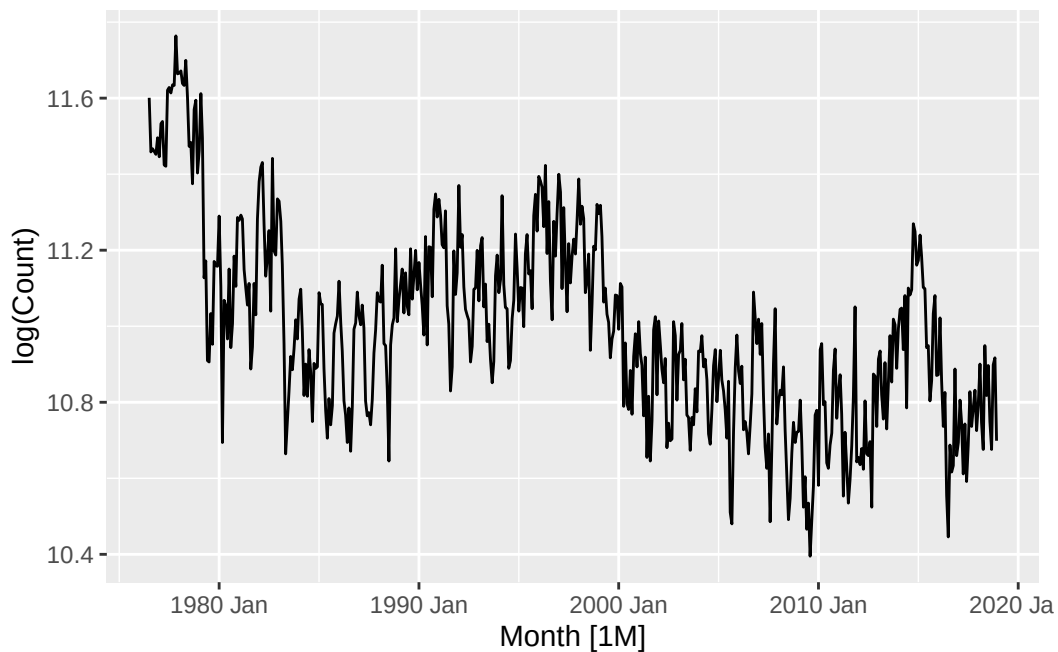
6 Bulls, bullocks an~ Vict~ stl    1976 Dec  98300 9.90e4      3765.    -
4441.
7 Bulls, bullocks an~ Vict~ stl    1977 Jan  93500 9.99e4      6980.    -
13369.
8 Bulls, bullocks an~ Vict~ stl    1977 Feb 102000 1.01e5      9526.    -
8402.
9 Bulls, bullocks an~ Vict~ stl    1977 Mar 102600 1.02e5      3375.    -
2637.
10 Bulls, bullocks an~ Vict~ stl    1977 Apr  91500 1.03e5     -4146.    -
7203.
# i 500 more rows
# i 1 more variable: season_adjust <dbl>

```

```

victoria_bulls |>
  autoplot(log(Count))

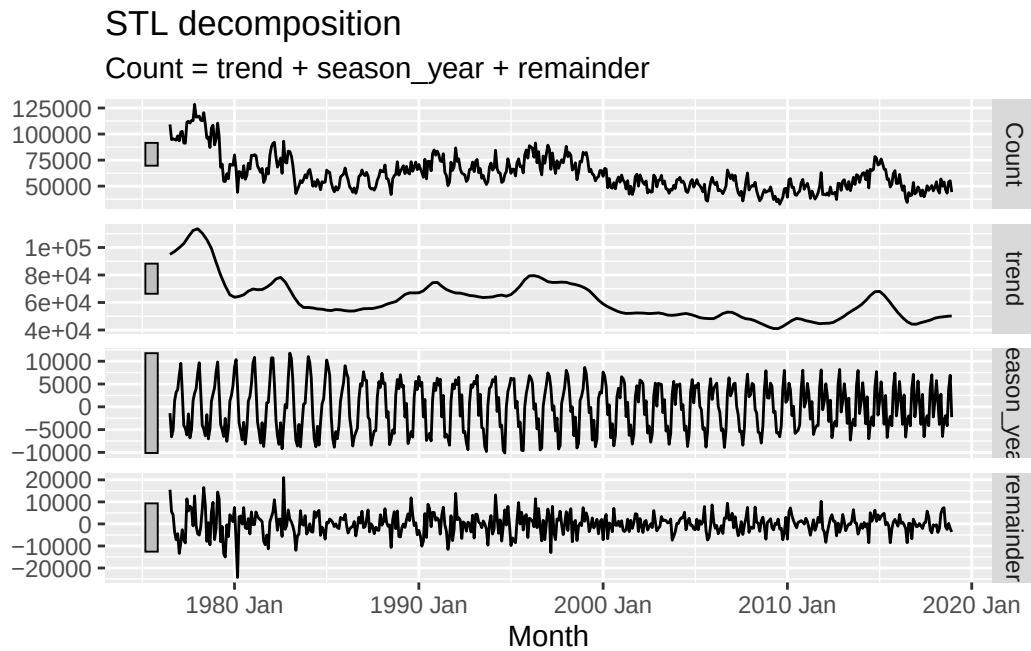
```



##Interpretation

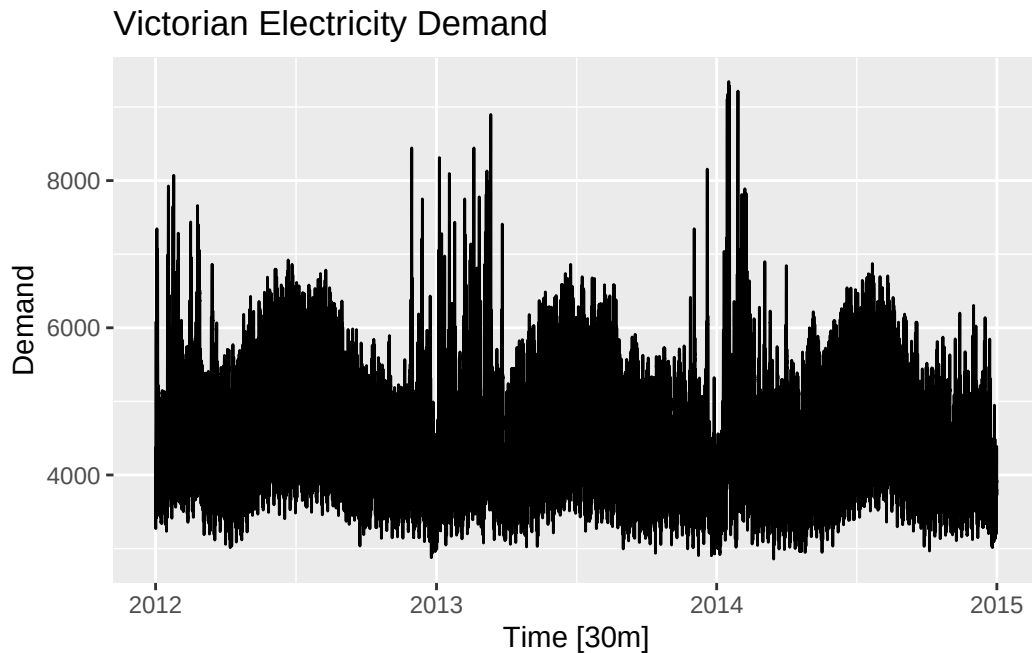
we could use a log transformation for aus_livestock observation over time, and by using a log transformation we see that the original series ranges differ from the transformed series ranges, but the overall pattern and variability remain similar to the original series.

```
components(dcmp) |> autoplot()
```



- Victorian Electricity Demand from `vic_elec`.

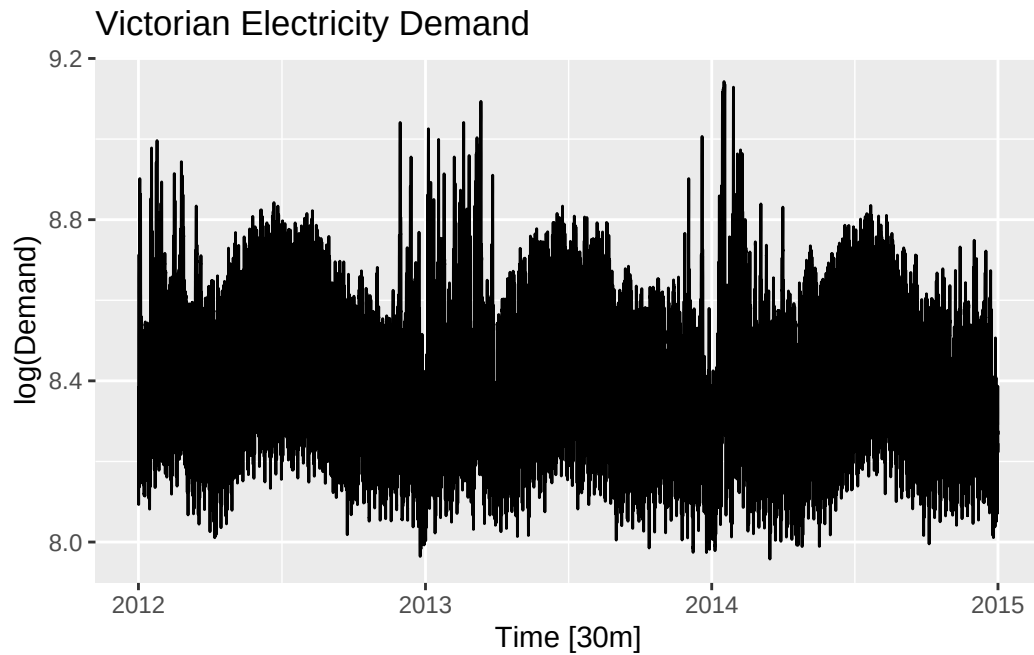
```
vic_elec |>  
  autoplot(Demand) + labs(title = "Victorian Electricity Demand")
```



Interpretation

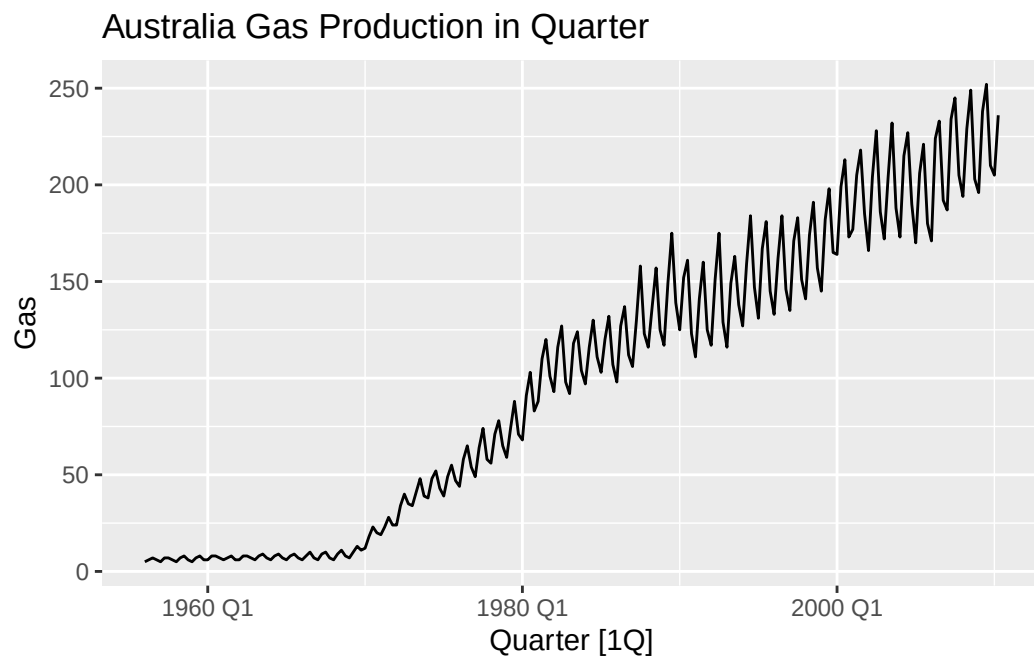
we could use a log transformation for `vic_elec` observation over time, and by using a log transformation we see that the original series ranges differ from the transformed series ranges, but the overall pattern and variability remain similar to the original series.

```
vic_elec |>
  autoplot(log(Demand)) + labs(title = "Victorian Electricity Demand")
```



- Gas production from `aus_production`.

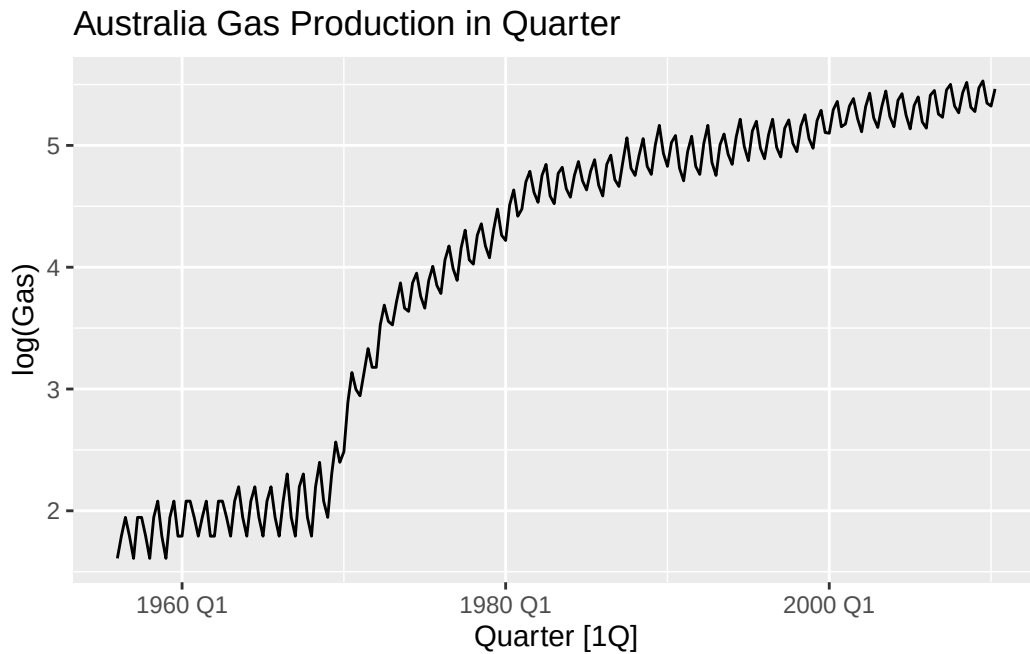
```
aus_production |>  
  autoplot(Gas) + labs(title = "Australia Gas Production in Quarter")
```



##Interpretation

The original gas production series shows an upward trend and seasonal fluctuations whose size changes with the level of the series. The log transformation compresses the larger values and makes the variation more stable over time.

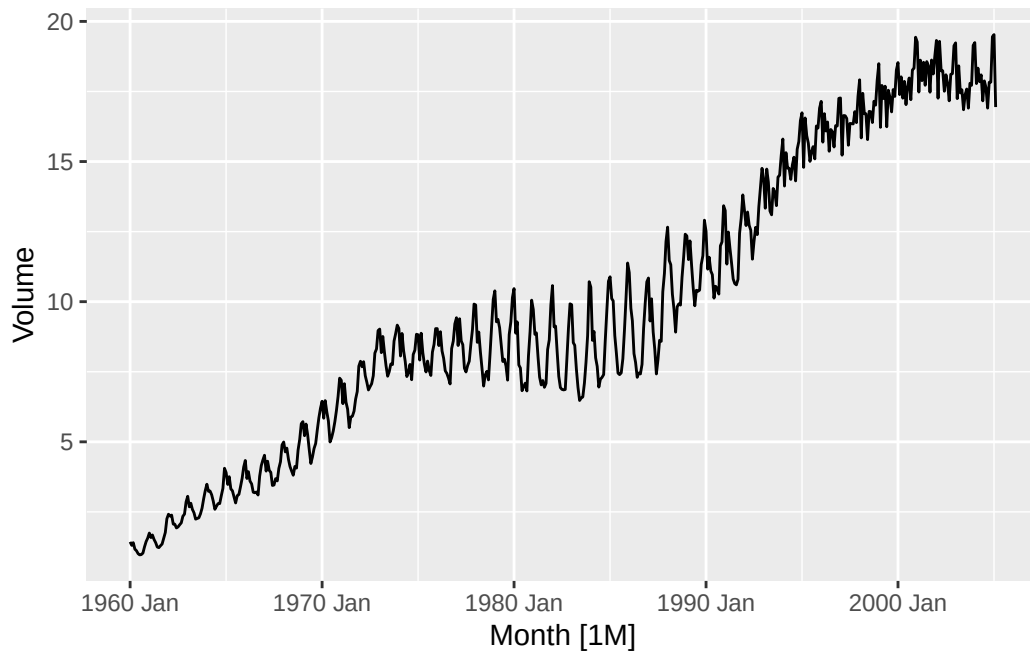
```
aus_production |>  
  autoplot(log(Gas)) + labs(title = "Australia Gas Production in Quarter")
```



3.3. Consider the last five years of the Gas data from aus_production.

let explore the canadian_gas data with a plot

```
canadian_gas |> autoplot(Volume)
```



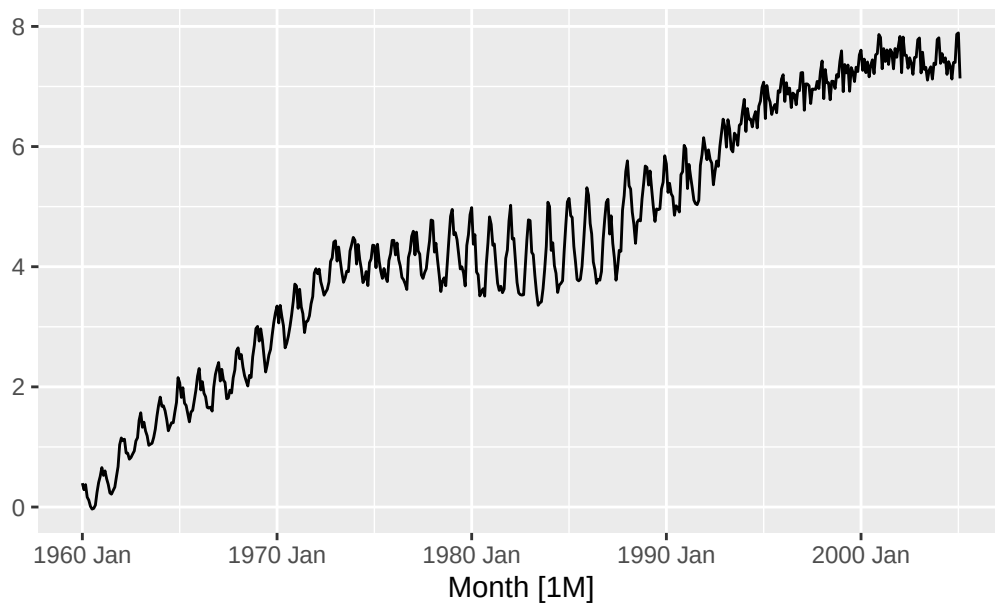
explore with Box - Cox

```
lambda <- canadian_gas |>
  features(Volume, features = guerrero) |>
  pull(lambda_guerrero)

canadian_gas |> autoplot(box_cox(Volume, lambda)) +
  labs(y = "",
       title = latex2exp::TeX(paste0( "Canada Transformed Gas Production with  $\lambda = "$ 

```

Canada Transformed Gas Production with $\lambda = 0.58$



Interpretation

Applying a Box–Cox transformation to the Canadian gas production data is not particularly helpful. Although the Guerrero method gives $\lambda = 0.58$ and the transformation changes the scale of the data, the overall pattern and variability remain similar to the original series. Therefore, the Box–Cox transformation does not provide a substantial improvement in stabilizing the variance.

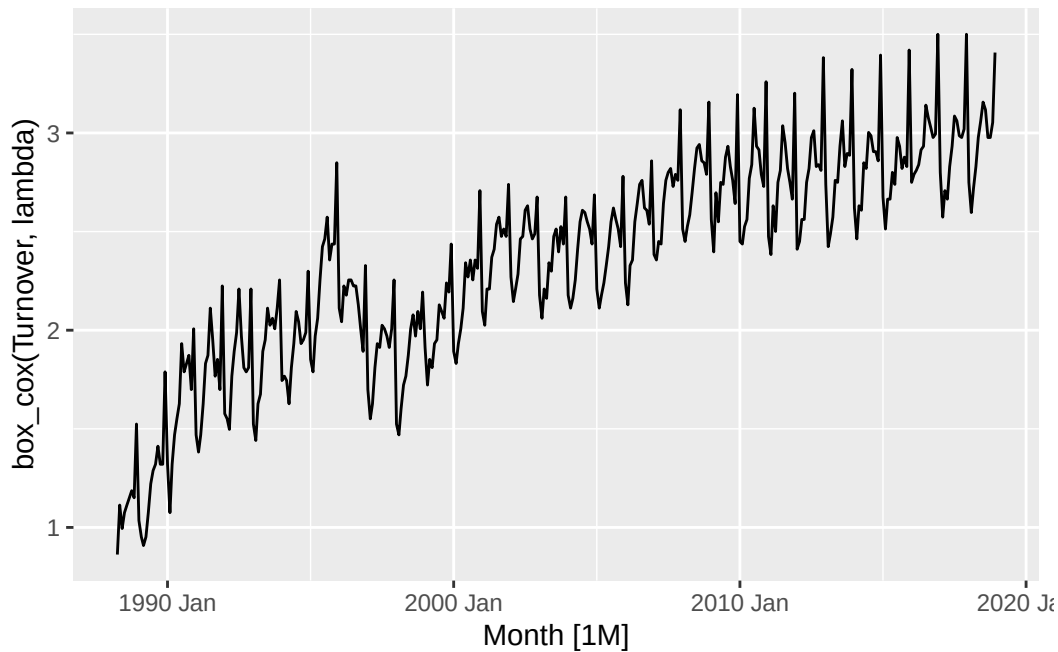
3.4. What Box-Cox transformation would you select for your retail data (from Exercise 7 in Section 2.10)?

```
set.seed(12345678)
myseries <- aus_retail |>
  filter(`Series ID` == sample(aus_retail$`Series ID`,1))

lambda <- myseries |>
  features(Turnover, features = guerrero) |>
  pull(lambda_guerrero)
lambda
```

```
[1] 0.08303631
```

```
myseries |> autoplot(box_cox(Turnover, lambda))
```



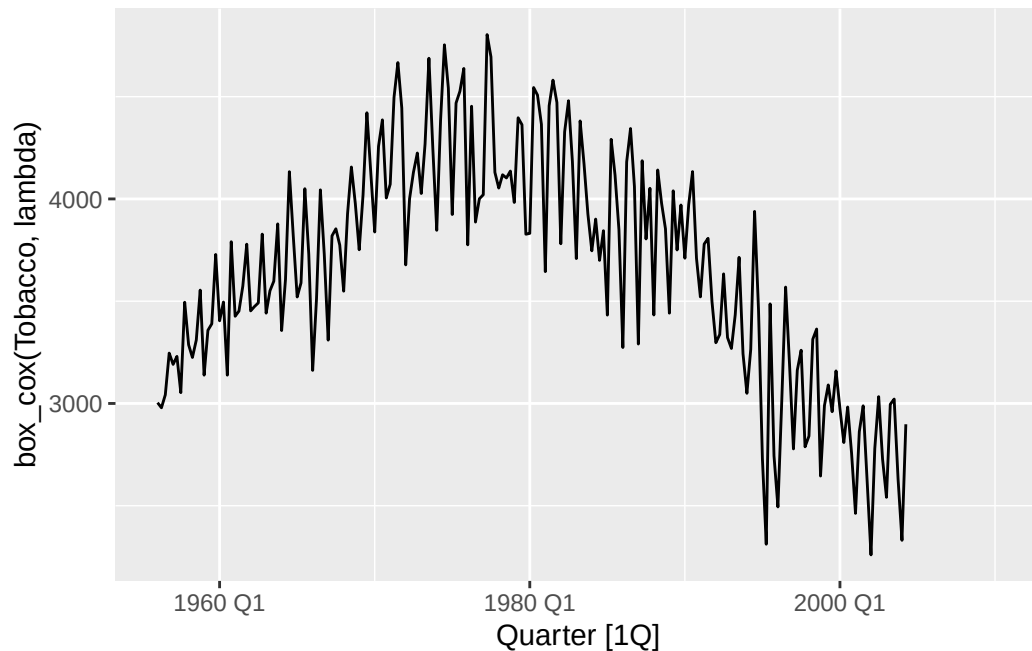
Interpretation

The box-cox I would select for my retail data from Exercise 7 in section 2.10 is automatically calculating the $\lambda = 0.083$ using features guerrero that will generate a lambda value we will use.

3.5. For the following series, find an appropriate Box-Cox transformation in order to stabilise the variance. Tobacco from aus_production, Economy class passengers between Melbourne and Sydney from ansett, and Pedestrian counts at Southern Cross Station from pedestrian.

```
lambda <- aus_production |>
  features(Tobacco, features = guerrero) |>
  pull(lambda_guerrero)

aus_production |> autoplot(box_cox(Tobacco, lambda), na.rm = TRUE)
```



Interpretation

To find an appropriate Box-Cox transformation in order to stabilise the variance, we use the features guerrero which provide a lambda that we apply to plot the graph therefor we can observe the graph shape for Australian tobacco production.

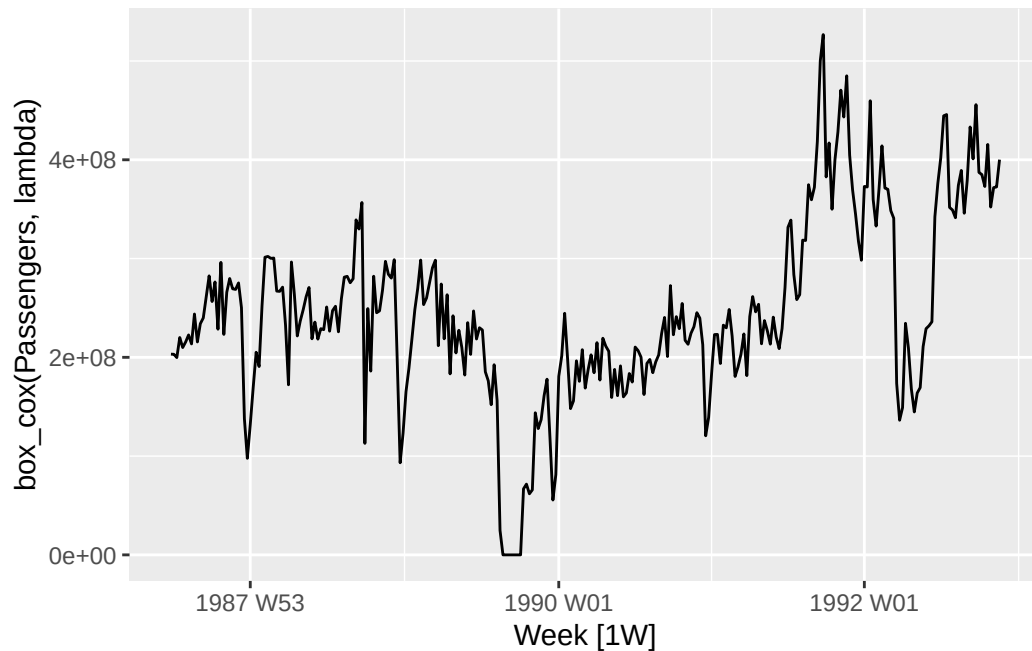
```
ansett_mel_sydney <- ansett |>
  filter(
    Class == "Economy",
    Airports == "MEL-SYD"
  )

lambda <- ansett_mel_sydney |>
  features(Passengers, features = guerrero) |>
  pull(lambda_guerrero)

lambda
```

```
[1] 1.999927
```

```
ansett_mel_sydney |>
  autoplot(box_cox(Passengers, lambda))
```



Interpretation

To find an appropriate Box-Cox transformation in order to stabilize the variance, we use the features guerrero which provide a lambda that we apply to plot the graph therefor we can observe the graph shape for ansett Economy flights. so we can concluded using box-cox is suitable.

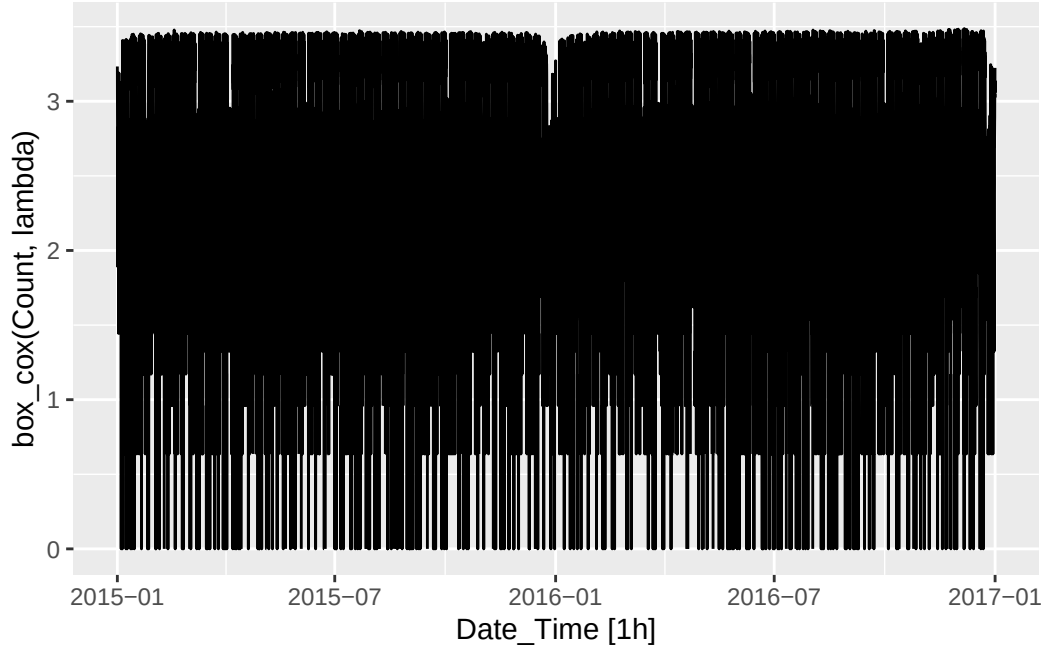
```
southern_cross <- pedestrian |>
  filter(Sensor == "Southern Cross Station")

lambda <- southern_cross |>
  features(Count, features = guerrero) |>
  pull(lambda_guerrero)

lambda
```

```
[1] -0.2501616
```

```
pedestrian |>
  filter(Sensor == "Southern Cross Station") |>
  autoplot(box_cox(Count, lambda))
```



##Interpretation

the series contains zero observations, making a standard Box-Cox transformation unsuitable without modifying the data.

3.6. Show that a 3×5 MA is equivalent to a 7-term weighted moving average with weights of 0.067, 0.133, 0.200, 0.200, 0.200, 0.133, and 0.067.

6. for 5-MA:

$$M_t = \frac{1}{5}(y_{t-2} + y_{t-1} + y_t + y_{t+1} + y_{t+2})$$

for 3x5 MA is:

$$T_t = \frac{1}{3}(M_{t-1} + M_t + M_{t+1})$$

for 5-MAs:

$$M_{t-1} = \frac{1}{5}(y_{t-3} + y_{t-2} + y_{t-1} + y_t + y_{t+1})$$

$$M_t = \frac{1}{5}(y_{t-2} + y_{t-1} + y_t + y_{t+1} + y_{t+2})$$

$$M_{t+1} = \frac{1}{5}(y_{t-1} + y_t + y_{t+1} + y_{t+2} + y_{t+3})$$

for Averaging 3 groups of 5 we have:

$$\frac{1}{3} \times \frac{1}{5} = \frac{1}{15} \approx 0.067$$

by counting the number of occur in each equation we get:

$$y_{t-3} = 1, y_{t-2} = 2, y_{t-1} = 3, y_t = 3, y_{t+1} = 3, y_{t+2} = 2, y_{t+3} = 1$$

each result divided by 15 we get:

$$T_t = 0.067y_{t-3} + 0.133y_{t-2} + 0.200y_{t-1} + 0.200y_t + 0.200y_{t+1} + 0.133y_{t+2} + 0.067y_{t+3}$$

3.7. Consider the last five years of the Gas data from aus_production.

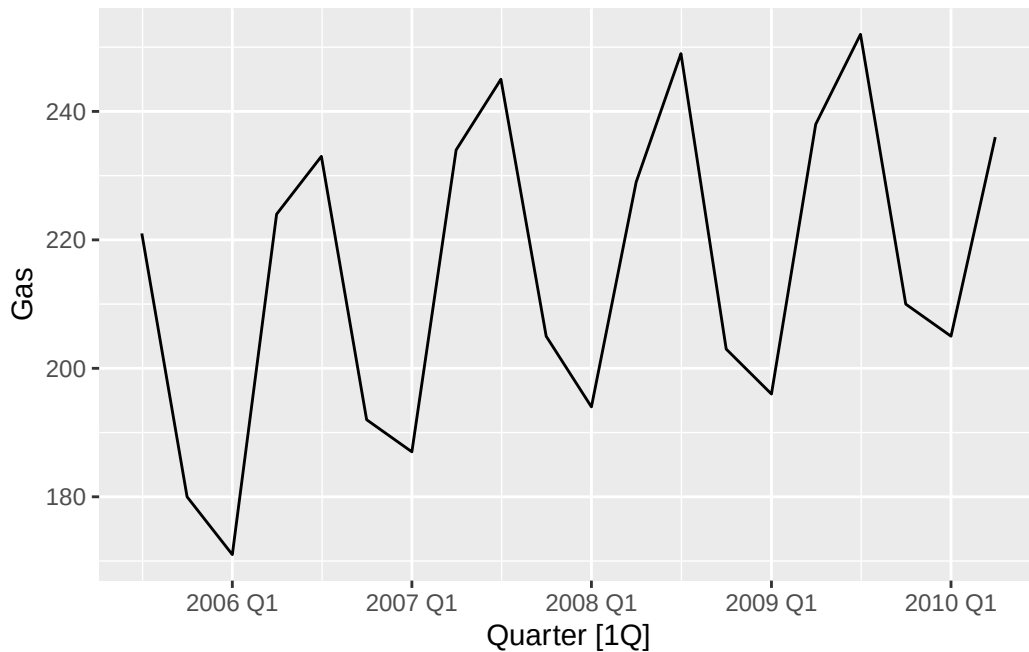
a. Plot the time series. Can you identify seasonal fluctuations and/or a trend-cycle?

```
gas <- tail(aus_production, 5*4) |> select(Gas)
gas
```

```
# A tsibble: 20 x 2 [1Q]
```

	Gas	Quarter
	<dbl>	<qtr>
1	221	2005 Q3
2	180	2005 Q4
3	171	2006 Q1
4	224	2006 Q2
5	233	2006 Q3
6	192	2006 Q4
7	187	2007 Q1
8	234	2007 Q2
9	245	2007 Q3
10	205	2007 Q4
11	194	2008 Q1
12	229	2008 Q2
13	249	2008 Q3
14	203	2008 Q4
15	196	2009 Q1
16	238	2009 Q2
17	252	2009 Q3
18	210	2009 Q4
19	205	2010 Q1
20	236	2010 Q2

```
gas |> autoplot(Gas)
```



Interpretation

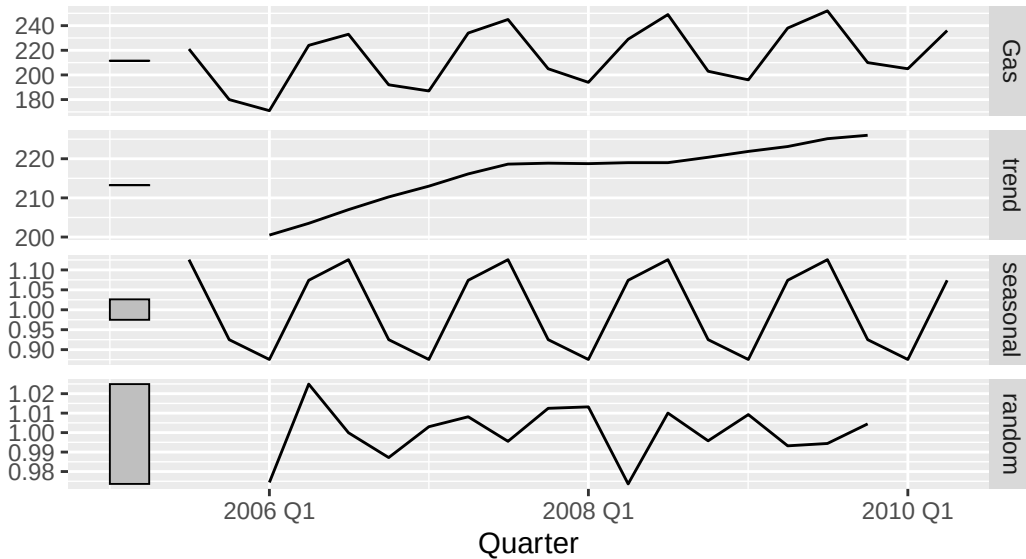
The series shows a repeating quarterly seasonal pattern over the five-year period. Gas production rises and falls at similar times each year, indicating strong seasonality. There is also an overall upward trend in gas production.

b. Use `classical_decomposition` with `type=multiplicative` to calculate the trend-cycle and seasonal indices.

```
gas |>
  model(classical_decomposition(Gas, type = "multiplicative")) |>
  components() |>
  autoplot(Gas, na.rm = TRUE) +
  labs(title = "Classical multiplicative decomposition of Gas Production in Australia")
```

Classical multiplicative decomposition of Gas Production in Aust

Gas = trend * seasonal * random



c. Do the results support the graphical interpretation from part a?

The results support the graphical interpretation from part (a). The classical decomposition shows a clear upward trend and a strong repeating quarterly seasonal pattern.

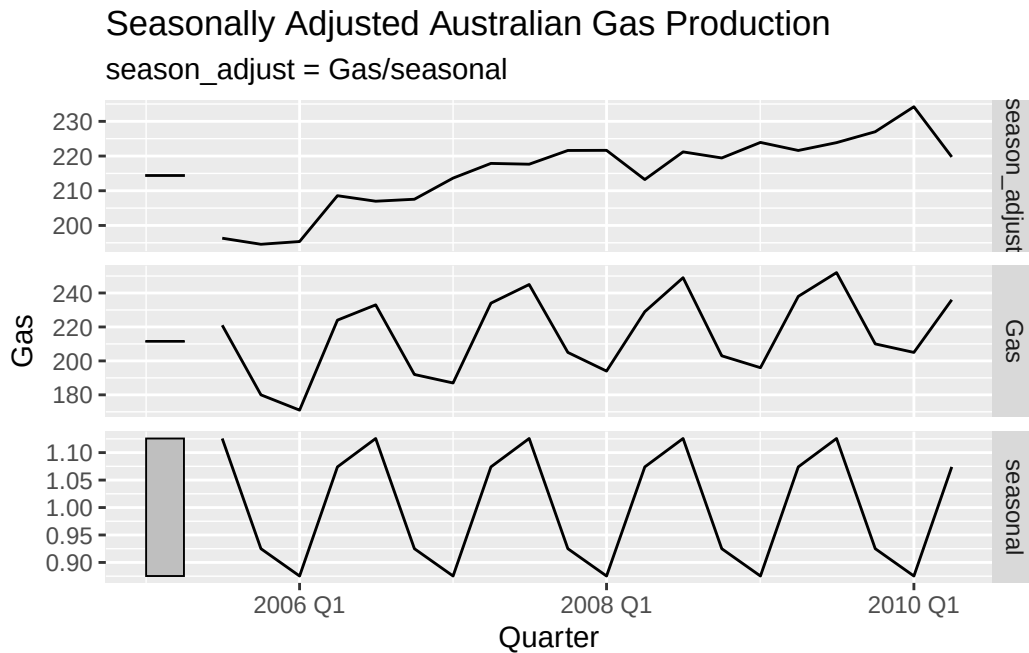
d. Compute and plot the seasonally adjusted data.

```
gas_dcmp <- gas |>
  model(
    classical_decomposition(Gas, type = "multiplicative")
  ) |>
  components()
gas_dcmp
```

```
# A dable: 20 x 7 [1Q]
# Key:      .model [1]
# :         Gas = trend * seasonal * random
#   .model      Quarter  Gas trend seasonal random season_adjust
#   <chr>         <qtr> <dbl> <dbl>    <dbl> <dbl>    <dbl>
# 1 "classical_decomposition(G~ 2005 Q3  221   NA    1.13   NA      196.
```

2	"classical_decomposition(G~	2005 Q4	180	NA	0.925	NA	195.
3	"classical_decomposition(G~	2006 Q1	171	200.	0.875	0.974	195.
4	"classical_decomposition(G~	2006 Q2	224	204.	1.07	1.02	209.
5	"classical_decomposition(G~	2006 Q3	233	207	1.13	1.000	207.
6	"classical_decomposition(G~	2006 Q4	192	210.	0.925	0.987	208.
7	"classical_decomposition(G~	2007 Q1	187	213	0.875	1.00	214.
8	"classical_decomposition(G~	2007 Q2	234	216.	1.07	1.01	218.
9	"classical_decomposition(G~	2007 Q3	245	219.	1.13	0.996	218.
10	"classical_decomposition(G~	2007 Q4	205	219.	0.925	1.01	222.
11	"classical_decomposition(G~	2008 Q1	194	219.	0.875	1.01	222.
12	"classical_decomposition(G~	2008 Q2	229	219	1.07	0.974	213.
13	"classical_decomposition(G~	2008 Q3	249	219	1.13	1.01	221.
14	"classical_decomposition(G~	2008 Q4	203	220.	0.925	0.996	219.
15	"classical_decomposition(G~	2009 Q1	196	222.	0.875	1.01	224.
16	"classical_decomposition(G~	2009 Q2	238	223.	1.07	0.993	222.
17	"classical_decomposition(G~	2009 Q3	252	225.	1.13	0.994	224.
18	"classical_decomposition(G~	2009 Q4	210	226	0.925	1.00	227.
19	"classical_decomposition(G~	2010 Q1	205	NA	0.875	NA	234.
20	"classical_decomposition(G~	2010 Q2	236	NA	1.07	NA	220.

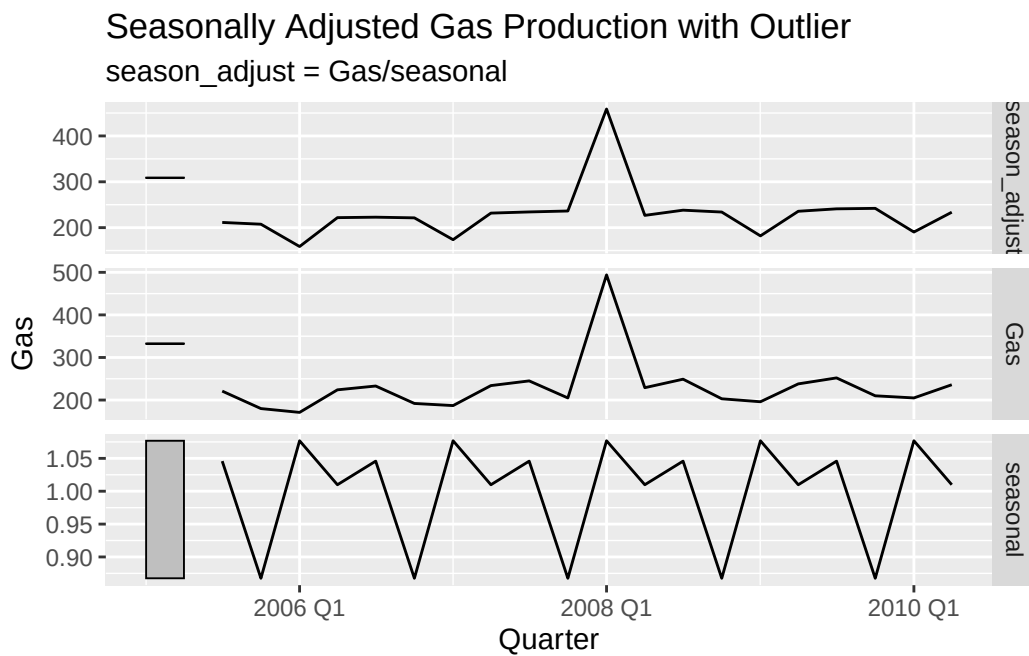
```
gas_dcmp |>
  autoplot(season_adjust) +
  labs(
    title = "Seasonally Adjusted Australian Gas Production",
    x = "Quarter",
    y = "Gas"
  )
```



e. Change one observation to be an outlier (e.g., add 300 to one observation), and recompute the seasonally adjusted data. What is the effect of the outlier?

```
gas_outlier <- gas |>
  mutate(
    Gas = if_else(
      Quarter == yearquarter("2008 Q1"),
      Gas + 300,
      Gas
    )
  )
gas_outlier_dcmp <- gas_outlier |>
  model(
    classical_decomposition(Gas, type = "multiplicative")
  ) |>
  components()
gas_outlier_dcmp |>
  autoplot(season_adjust) +
  labs(
    title = "Seasonally Adjusted Gas Production with Outlier",
    x = "Quarter",
```

```
y = "Gas"
)
```



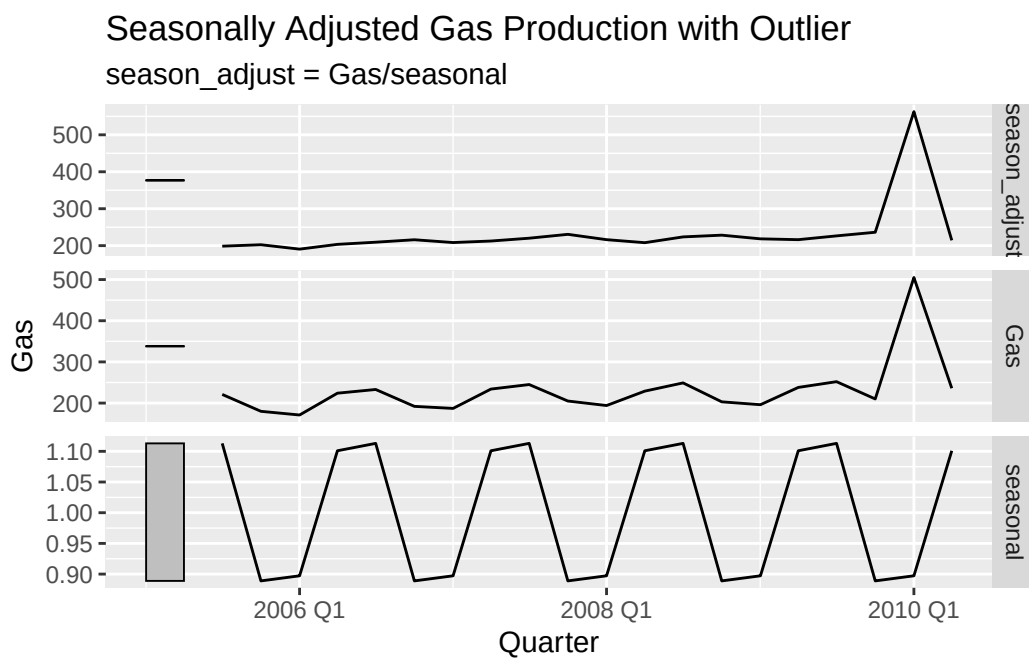
Interpretation

Adding 300 to one observation creates a large spike in the seasonally adjusted data. The outlier has a strong effect on that observation and can also affect the estimated components used in the classical decomposition.

f. Does it make any difference if the outlier is near the end rather than in the middle of the time series?

```
gas_outlier <- gas |>
mutate(
  Gas = if_else(
    Quarter == yearquarter("2010 Q1"),
    Gas + 300,
    Gas
  )
)
```

```
gas_outlier_dcmp <- gas_outlier |>
  model(
    classical_decomposition(Gas, type = "multiplicative")
  ) |>
  components()
gas_outlier_dcmp |>
  autoplot(season_adjust) +
  labs(
    title = "Seasonally Adjusted Gas Production with Outlier",
    x = "Quarter",
    y = "Gas"
  )
)
```



Interpretation

Yes, it makes a difference when the outlier is near the end rather than in the middle of the time series. An outlier in the middle can affect the estimated trend-cycle around it because there are observations on both sides. Near the end, the trend-cycle cannot be estimated in the same way because there are not enough observations after the outlier. Therefore, the position of the outlier affects the classical decomposition and the seasonally adjusted data.

3.8. Recall your retail time series data (from Exercise 7 in Section 2.10). Decompose the series using X-11. Does it reveal any outliers, or unusual features that you had not noticed previously?

```
#install.packages("seasonal")
library(seasonal)
```

Warning: package 'seasonal' was built under R version 4.5.2

Attaching package: 'seasonal'

The following object is masked from 'package:tibble':

view

```
set.seed(12345678)
myseries <- aus_retail |>
  filter(`Series ID` == sample(aus_retail$`Series ID`,1))
myseries
```

A tsibble: 369 x 5 [1M]

Key: State, Industry [1]

	State	Industry	`Series ID`	Month	Turnover
	<chr>	<chr>	<chr>	<moth>	<dbl>
1	Northern Territory	Clothing, footwear and pers~	A3349767W	1988 Apr	2.3
2	Northern Territory	Clothing, footwear and pers~	A3349767W	1988 May	2.9
3	Northern Territory	Clothing, footwear and pers~	A3349767W	1988 Jun	2.6
4	Northern Territory	Clothing, footwear and pers~	A3349767W	1988 Jul	2.8
5	Northern Territory	Clothing, footwear and pers~	A3349767W	1988 Aug	2.9
6	Northern Territory	Clothing, footwear and pers~	A3349767W	1988 Sep	3
7	Northern Territory	Clothing, footwear and pers~	A3349767W	1988 Oct	3.1
8	Northern Territory	Clothing, footwear and pers~	A3349767W	1988 Nov	3
9	Northern Territory	Clothing, footwear and pers~	A3349767W	1988 Dec	4.2
10	Northern Territory	Clothing, footwear and pers~	A3349767W	1989 Jan	2.7

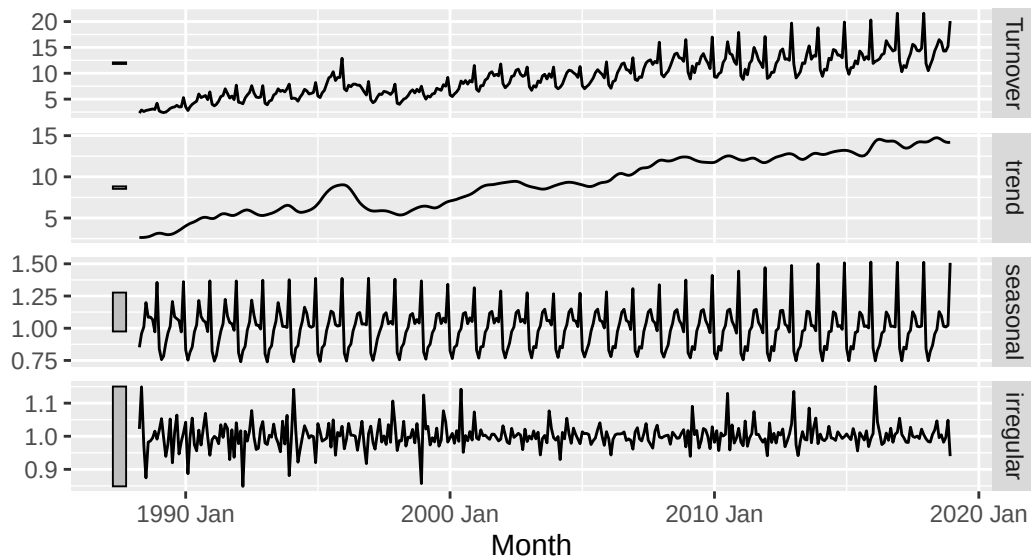
i 359 more rows

```
retail_outlier_dcmp <- myseries |>
  model(
    x11 = X_13ARIMA_SEATS(Turnover ~ x11())
  )

retail_outlier_dcmp |>
  components() |>
  autoplot()
```

X-13ARIMA-SEATS using X-11 adjustment decomposition

Turnover = trend * seasonal * irregular



Interpretation

Observing the remainder component, we can see some unusual positive and negative spikes that were not as noticeable in the original time series. These large deviations represent observations that are not explained by the trend or seasonal components and may indicate unusual events or outliers.

3.9. Figures 3.19 and 3.20 show the result of decomposing the number of persons in the civilian labor force in Australia each month from February 1978 to August 1995.

a. Write about 3–5 sentences describing the results of the decomposition. Pay particular attention to the scales of the graphs in making your interpretation.

The decomposition shows a smooth upward trend in the Australian civilian labor force from 1978 to 1995. The seasonal component ranges approximately from -100 to $+100$ and shows a consistent seasonal pattern. December and March have relatively large positive seasonal effects, while January and August have large negative seasonal effects. Compared with the overall level of the series, which increases from approximately 6,500 to 9,000, the seasonal fluctuations are relatively small.

b. Is the recession of 1991/1992 visible in the estimated components?

Yes, the recession of 1991/1992 is visible in the decomposition. The trend component shows that the growth of the labor force slows considerably around 1991–1992. The remainder also shows unusually large negative values around 1991, including a decline of approximately -400 .